

Founder's Research Track Record & Scientific Foundation

T4 · Research evidence · Jinkyoo Park · KAIST · Six selected co-authored papers

Jinkyoo Park · Omelet co-founder and CEO · KAIST faculty since 2016. The [official KAIST profile](#) lists his Industrial & Systems Engineering appointment and research in neural combinatorial optimization, multi-agent reinforcement learning and industrial design optimization.

Six selected co-authored publications are linked below to primary publication records, accessible papers and five author-linked code repositories. They document research expertise relevant to Omelet's technical direction.

Attribution. Jinkyoo Park appears in the author list of every selected paper. These are collaborative academic publications. The relevance notes are Omelet's interpretation of the research connection, not evidence that each method is deployed in a customer product or that the company exclusively owns the research.

RL4CO KDD 2025

RL4CO: An Extensive Reinforcement Learning for Combinatorial Optimization Benchmark

Federico Berto et al.; Jinkyoo Park is a listed co-author.

Contribution: A shared research framework spanning 27 combinatorial-optimization environments and 23 baselines, with modular training and evaluation.

Relevance: Reusable optimization infrastructure and systematic comparison of learned solvers.

[Publication / author record](#) · [Read paper](#) · [Source code](#)

The author-maintained arXiv record identifies KDD 2025 Oral and links the ACM DOI. Authorship and the 27/23 counts are visible in the paper.

PARCO NeurIPS 2025

PARCO: Parallel AutoRegressive Models for Multi-Agent Combinatorial Optimization

Federico Berto, Chuanbo Hua, Laurin Luttmann, Jiwoo Son, Junyoung Park, Kyuree Ahn, Changhyun Kwon, Lin Xie, Jinkyoo Park.

Contribution: Parallel solution construction with agent communication and conflict handling for multi-agent routing and scheduling.

Relevance: Coordinated decision-making across multiple vehicles or resources.

[Publication / author record](#) · [Read paper](#) · [Source code](#)

NeurIPS proceedings confirm the conference year and Jinkyoo Park in the author list.

Sym-NCO NeurIPS 2022

Sym-NCO: Leveraging Symmetricity for Neural Combinatorial Optimization

Minsu Kim, Junyoung Park, Jinkyoo Park.

Contribution: Symmetry-based reinforcement-learning regularization evaluated across four routing-related optimization tasks.

Relevance: Generalization across problem representations and learning efficient optimization policies.

[Publication / author record](#) · [Read paper](#) · [Source code](#)

NeurIPS proceedings confirm the conference year, authors and author-linked source code.

Selected publications · Scaling, routing and design

Meta-SAGE ICML 2023

Meta-SAGE: Scale Meta-Learning Scheduled Adaptation with Guided Exploration for Mitigating Scale Shift on Combinatorial Optimization

Jiwoo Son, Minsu Kim, Hyeonah Kim, Jinkyoo Park.

Contribution: Adapts pretrained optimization policies to larger instances using scale-aware representations and guided test-time exploration.

Relevance: Research on scaling learned solvers beyond their training-instance sizes.

[Publication / author record](#) · [Read paper](#) · [Source code](#)

PMLR confirms ICML 2023 and all four authors; the abstract links the source repository.

Neuro CROSS-exchange ICLR 2023

Learning to CROSS exchange to solve min-max vehicle routing problems

Minjun Kim, Junyoung Park, Jinkyoo Park.

Contribution: A graph neural network predicts promising route exchanges to guide a conventional search operator for min-max vehicle routing.

Relevance: Hybrid AI and operations-research methods for practical routing decisions.

[Publication / author record](#) · [Read paper](#) · [Conference slides](#)

ICLR confirms authorship and the accepted title above. The linked earlier preprint uses the title “Neuro CROSS exchange: Learning to CROSS exchange to solve realistic vehicle routing problems”.

DevFormer ICML 2023

DevFormer: A Symmetric Transformer for Context-Aware Device Placement

Haeyeon Kim, Minsu Kim, Federico Berto, Joungho Kim, Jinkyoo Park.

Contribution: Context-aware transformer optimization for decoupling-capacitor placement, evaluated in simulated and real hardware.

Relevance: Evidence of optimization research breadth beyond logistics, including semiconductor-related design.

[Publication / author record](#) · [Read paper](#) · [Source code](#)

PMLR confirms ICML 2023 and the author list. The author repository provides code and a placement benchmark.

Pitch claims, portfolio scope and sources

Official lab-list recount, 17 September 2026: 61 entries under “ML Conference Papers”, including 36 venue lines matching NeurIPS / ICML / ICLR. Jinkyoo Park is named in the lab’s author lines for all 61 entries. Six selected papers above were additionally checked against primary author/publication records. These counts describe the academic portfolio, not 61 company product papers or a count of deployed algorithms.

CLAIM	DIRECT SOURCE	SCOPE
RL4CO, KDD 2025; 27 environments; 23 baselines	Publication DOI · Open full paper	Publication and counts supported by the paper and KAIST publication record. The author-maintained arXiv record identifies “KDD 2025 Oral”; see the linked publication record above.
Learned-hybrid ~250× faster at 7K nodes	Paper Table 5.4	LKH-3: 501 s, objective 245.0; GLOP-G (AM), RL4CO implementation: 2.0 s, objective 190.9. 501 / 2.0 = 250.5×. Published research comparison, not an Omelet customer-product or cuOpt measurement.
61 ML conference papers; 36 at NeurIPS / ICML / ICLR	KAIST SILAB publication list	Checked 17 September 2026: 61 entries in the ML Conference Papers list; 36 venue lines match the three named conferences. Lab publication history, not a count of company-authored product papers.
897 RL4CO stars in the pitch	Public repository · Dated supplied screenshot	The supplied 14 September screenshot shows 900 stars / 153 forks. Counts change; it does not establish the exact earlier 897 snapshot.

Research versus commercial evidence

The public paper and repository support the research foundation and its authorship. They do not establish the deployment state of Omelet’s commercial platform or NVIDIA components at any named customer. The earlier NeurIPS 2023 workshop version and the KDD 2025 publication are distinguished; the KDD Oral label is attributed to the author-maintained arXiv record.

Publication count method

Count direct entries in the lab page’s ML Conference Papers list; then count their venue lines containing NeurIPS, ICML or ICLR. Workshop and engineering-paper lists are excluded. The live list may change after this dated review.

[Open dated publication-count register \(CSV\)](#)

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